

Project 4:

Run the Bert Model

I. Source Code Download:

The two source code files for this project can be accessed on bb.

II. The Bert Model

1. Training Task (Your project should run both of them.)

(a) Single Sentence Classification Dataset: **sst2**

The Stanford Sentiment Treebank is a binary single-sentence classification task consisting of sentences extracted from movie reviews with human annotations of their sentiment, 0 means positive while 1 means negative. Some examples are as follows.

it 's a charming and often affecting journey .	1 positive
unflinchingly bleak and desperate	0 negative
allows us to hope that nolan is poised to embark a major career as a commercial yet inventive filmmaker .	1 positive
the acting , costumes , music , cinematography and sound are all astounding given the production 's austere locales .	1 positive
it 's slow -- very , very slow .	0 negative

(b) Sentence Pair Classification Dataset: **mrpc**

Microsoft Research Paraphrase Corpus consists of sentence pairs automatically extracted from online news sources, with human annotations for whether the sentences in the pair are semantically equivalent, 1 means equivalent while 0 means not. Some examples are as follows.

He said the foodservice pie business doesn 't fit the company 's long-term growth strategy .	" The foodservice pie business does not fit our long-term growth strategy .	1 equivalent
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The AFL-CIO is waiting until October to decide if it will endorse a candidate .	The AFL-CIO announced Wednesday that it will decide in October whether to endorse a candidate before the primaries .	1 equivalent
No dates have been set for the civil or the criminal trial .	No dates have been set for the criminal or civil cases , but Shanley has pleaded not guilty .	0 not_equivalent

2. Running the Model

Before running, you should download two libraries called **transformers** and **datasets** using pip command:

pip install transformers

pip install datasets

The two Python files are named after the tasks and serve as the runner of the Bert model on their corresponding tasks. They will automatically download the task datasets and a small pretrained Bert model with 4 encoder layers from the internet, finetune the model for 2 epochs on the training set, and evaluate its performance (accuracy for mrpc, accuracy, and F1 for sst2) on the evaluation set at the end of each epoch. The progress bar on the command line shows the progress of the whole training process, not a single epoch. The model will be saved to the local path after training. Finally, 5 practical examples are provided where the trained model is used to solve the task.

Please include the performance after the final epoch on the evaluation set and the screenshot of the practical examples in your report, as no test set is available due to no authorization. Also note that the training set of mrpc is rather small compared to that of sst2, so it should be the first choice for you to run to examine the program's functionality.